EXPERIMENT 02

Finite State Automaton for Strings Ending with 'ab'

Aim

To implement a basic finite-state automaton that accepts strings ending with the pattern 'ab'.

Procedure

1. Define states representing the progress toward the ending pattern.
2. Read each character and update the state.
3. Accept the string if the final state represents a suffix 'ab'.
4. Test the automaton with sample strings.

Program

```
def ends_with_ab(text):
    state = 0
    for ch in text:
        if state == 0:
            state = 1 if ch == "a" else 0
        elif state == 1:
            state = 2 if ch == "b" else (1 if ch == "a" else 0)
        else:  # current suffix is "ab"
            state = 1 if ch == "a" else 0
    return state == 2

for value in ["ab", "aab", "abab", "aba", "abb"]:
    print(value, "->", "Accepted" if ends_with_ab(value) else "Rejected")
```

Output

```
OUTPUT — EXPERIMENT 02
ab -> Accepted
aab -> Accepted
abab -> Accepted
aba -> Rejected
abb -> Rejected
```

Result

Thus, the finite-state automaton accepted strings ending with 'ab' and rejected the others.